

Airbnb Bookings Predictions

SVM and Neural Network Model Performance Analysis

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Claire Tours, a UK-based touring company, would like to partner with Airbnbs in Cape Town to sell touring packages. They are conducting research to predict which potential hosts would be ideal candidates to partner with based on their attributes. They thus require predictions of demand/vacancy at these properties. In this case, demand is proxied by the target variable, `fully_booked_30`, which is a binary variable that takes the value 1 if the Airbnb is fully booked in the next 30 days and 0 otherwise. This is done by training 3 classification models and comparing their performance metrics. The models are:

- SVM with a linear kernel
- SVM with a radial kernel
- Feed-forward Neural Network

The predictions from the best-performing model, which can accurately classify Airbnbs that will not be fully booked in the next 30 days, will serve as input for the company's potential monthly revenue models.

10-fold cross-validation was used to train all the models, each with its own set of hyperparameters. The summary of the best-performing model for each is then selected and presented, along with a plot of the average CV metric across hyperparameter combinations.

Exploratory Analysis

The data is highly imbalanced in favour of the negative class) with ~82% of observations belonging to `fully_booked_30=no`. Since the modelling aims to predict availability, this will be treated as the positive class in modelling- this is the target class.

There is no clear distinction between the classes spatially with both classes spread out evenly across latitudes and longitudes.

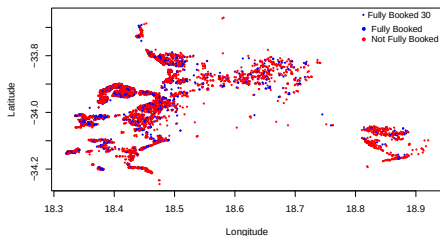


Figure 1: Spatial Distribution of Listings Grouped by Target

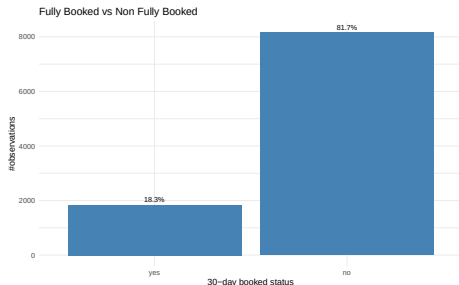
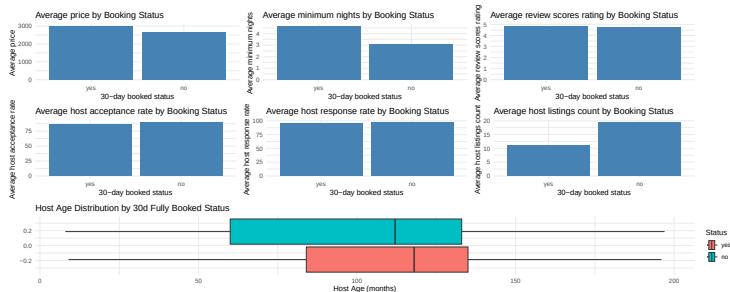


Figure 2: Distribution of Target, `fully_booked_30`

Exploratory Analysis Feature Distributions

When grouping by the target variable, `fully_booked_30`, the following key insights are observed:

- Average Price and average minimum nights are slightly higher for observations in the “yes” class, with the gap much wider for average minimum nights (~ 50 % higher than the “no” class).
- Average review scores and average response rate are fairly evenly split across the two target classes
- Average host listings are higher for the “no” class, with an average of ~20 listings.
- For the host age (measured by how long the host id has been an Airbnb in months) the median (118 months) host age is ~5% higher for the “yes” class compared to the no class. Moreover, the no class has a longer range of host ages, although the distributions about the median for both groups are similar; most host ages lie below their respective median host age.



Hyperparameter definitions

Cost Parameter: This is a constant >0 controlling how strict the model (alternatively, how wide or narrow the margin is around the linear decision boundary). A higher cost value indicates a stricter model (and thus a narrower margin, with fewer boundary violations), whilst a lower cost indicates a more tolerant model (or wider margin). For the linear kernel SVM, the cost parameter is tuned with the following options:

1. Sequence starting from 0.15 to 3, in steps of 0.15, amounting to 20 cost parameter options.

Table 1: Linear Kernel SVM Best Model Summary

Metric	Value
best_cost	0.15
nr.support_vectors	3807
best_cross_validation_error	0.18
best_training_accuracy	0.82

Linear Kernel SVM Gridsearch Summary

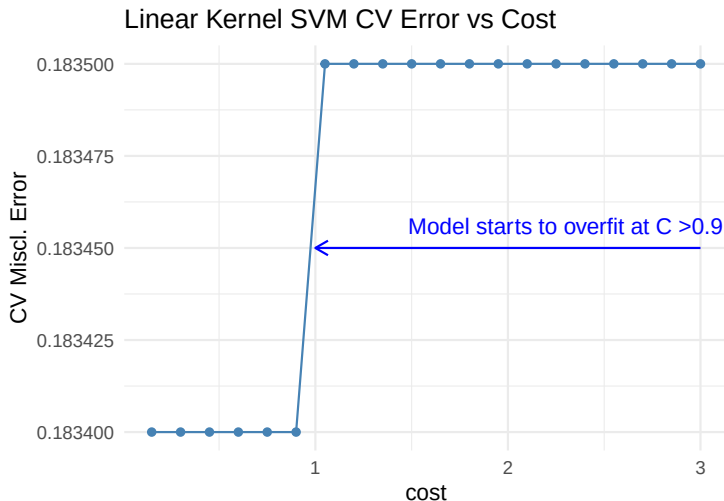


Figure 4: Linear Kernel SVM CV Error vs Cost

Non-Linear SVM Tuning Summary

A radial-kernel SVM was chosen to train on the data. This kernel has an additional similarity parameter, gamma, that controls the rate of (dis)similarity between points based on their distance; thus, it is also tuned during training in addition to the cost parameter:

- ① Cost: Sequence starting from 0.15 to 3, in steps of 0.15, amounting to 20 cost parameter options
- ② Gamma: [0.05 0.25 0.45 0.65 0.85 1.00]

Table 2: Radial Kernel SVM Best Model summary

Metric	Value
nr.support_vectors	7421.00
best_cost	0.75
best_gamma	0.45
best_cross_validation_error	0.18
best_training_accuracy	0.85

Model Training Non-Linear Kernel SVM

Gridsearch Training Summary

Across parameter combinations, the model starts to overfit as the gamma parameter increases past 0.50

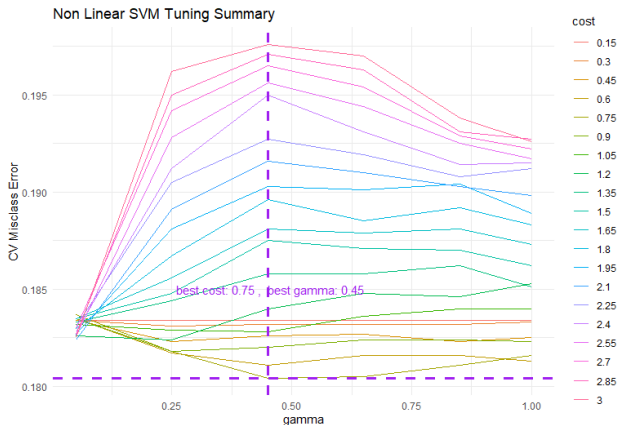


Figure 5: Non-linear SVM CV Misc. Error per Grid Hyperparameter Combination

Hyperparameters

For the feed-forward neural net, three hyperparameters were tuned in the training process:

- ① **Number of hidden layer nodes:** [2, 5, 10,15]
- ② **L1 regularization parameter**, which helps hedge the model against overfitting by minimizing (to 0 for some) the weights coming from non-relevant nodes (including the input layer). Values of this control the strength of the penalty, with smaller values indicating a more flexible and tolerant model: [0.5,0.15,0.25]
- ③ **Number of Training Epochs**- the number of times the (entire) training data feeds through the neural net: [500, 1000, 1500,2000], although early stopping is enabled to prevent the model from overfitting.

The candidate model with the lowest CV log loss is chosen.

Best Model Summary

Table 3: Neural Net Best Model Metrics

Metric	Value
best L1	0.05
best nr. hidden layer nodes	2.00
best CV Accuracy	0.82
best CV Logloss	0.45

Gridsearch Summary

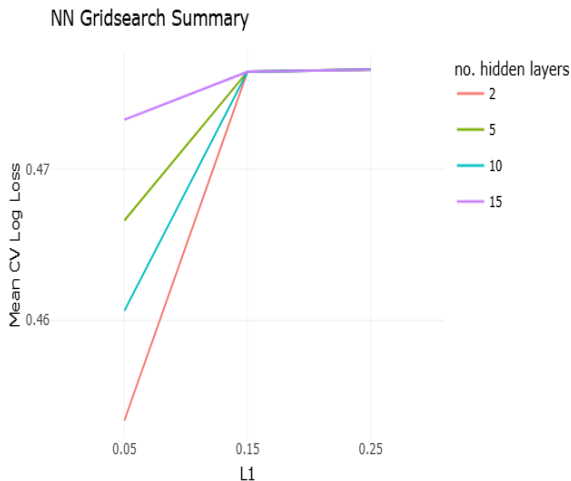


Figure 6: NN CV Log Loss Per Grid Hyperparameter Combination

Cross Model Comparison

Key Insights

- Both SVMs show perfect sensitivity scores, that is, they are able to perfectly predict all positive labels as positive (target=no); however, they had very poor specificity. The models were not able to correctly predict the majority of the negative class. The F1 scores for both are relatively high due to perfect positive class predictions, whilst the balanced accuracy suffers due to both models' inability to predict the minority class.
- The NN model is slightly more balanced in its metrics, at different probability class assignment thresholds:
 - At $\tau = 0.51$, the model was able to correctly classify ~100% the target class (fully_booked=no), with zero correct minority class predictions. The F1 score is also high, mostly bolstered by the high positive class prediction accuracy; the balanced accuracy score is at 50%- the model performs no better than a coin toss (again, mostly due to poor correct minority class predictions).
 - At $\tau = 0.81$, the model was able to correctly classify ~73% of the minority (specificity) class label and only ~51% of the majority, positive label class. The F1 score decreases to 0.65, although balanced accuracy improves to 65% due to the improvement in correct negative (minority class) predictions.

Table 4: Cross Model CV Metrics

Model	Specificity	Sensitivity	F1	Bal.Accuracy
Linear Kernel SVM	0.000	1.000	0.899	0.500
Radial Kernel SVM	0.195	1.000	0.917	0.598
Feed Forward Neural Net (=	0.002	0.999	0.899	0.500